

OSLAB SKILL-10 PROGRAMS & OUTPUTS

Date: 04-10-2026

Program-1 Code & Output:

```
sainath@Sainath:~/OSLAB/skill10$ gcc program1.c -o program1
sainath@Sainath:~/OSLAB/skill10$ cat program1.c
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main() {
    char dir[200];
    char cmd[20];
    FILE *fp;

    while (1) {
        printf("ossh> ");
        scanf("%19s", cmd);

        if (strcmp(cmd, "pwd") == 0) {
            getcwd(dir, sizeof(dir));
            printf("%s\n", dir);
        }
        else if (strcmp(cmd, "exit") == 0) {
            getcwd(dir, sizeof(dir));

            fp = fopen("state.txt", "w");
            fprintf(fp, "%s\n", dir);
            fclose(fp);

            printf("State saved. Exiting...\n");
            break;
        }
        else {
            printf("Invalid command\n");
        }
    }

    return 0;
}

sainath@Sainath:~/OSLAB/skill10$ ./program1
ossh> pwd
/home/sainath/OSLAB/skill10
ossh> exit
State saved. Exiting...
```

Program-2 Code & Output:

```
sainath@Sainath:~/OSLAB/skill10$ cat program2.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <sys/wait.h>

int valid(char *name) {
    if (!isalpha(name[0]) && name[0] != '_')
        return 0;

    for (int i = 1; name[i]; i++)
        if (!isalnum(name[i]) && name[i] != '_')
            return 0;

    return 1;
}

int main() {
    char input[100], *name, *value;

    printf("Enter export command: ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    if (strncmp(input, "export ", 7) != 0) {
        printf("Invalid export syntax\n");
        return 1;
    }

    name = strtok(input + 7, "=");
    value = strtok(NULL, "");

    if (!name || !value || !valid(name)) {
        printf("Invalid variable name or syntax\n");
        return 1;
    }

    setenv(name, value, 1);
    printf("%s=%s\n", name, getenv(name));

    if (fork() == 0) {
        printf("Child: %s=%s\n", name, getenv(name));
        exit(0);
    }

    wait(NULL);
    return 0;
}
```

```
sainath@Sainath:~/OSLAB/skill10$ ./program2
Enter export command: export CITY=Hyderabad
CITY=Hyderabad
Child: CITY=Hyderabad
```

History:

```
692 cd OSLAB
693 ls
694 mkdir skill10
695 cd skill10
696 gedit program1.c
697 gcc program1.c -o program1
698 cat program1.c
699 ./program1
700 gedit program2.c
701 gcc program2.c -o program2
702 cat program2.c
703 ./program2
704 history
```